

Bowen Yu

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Education

Massachusetts Institute of Technology (*B.S. & MEng, Physics & AI*) Aug. 2024 — Jun. 2028
Peking University (*B.S., Physics*) Sep. 2023 — Jul. 2024
No.1 High School Affiliated to Central China Normal Univ. Sep. 2020 — Jul. 2023

- GPA: 5.00/5.00 (*MIT*); 3.91/4.00 (*Peking Univ.*)
- Selected Coursework: Machine Learning, Deep Learning, Natural Language Processing, Discrete Probability and Stochastic Processes, Computational Architecture, Quantum Physics I, II & III

Working Experiences

Quantitative Research Analyst @ Citadel Securities (*Miami, FL, United States*) Jun. 2026 — Present

- Ongoing summer quantitative research intern in systematic equities

AI & Material Science Research Assistant @ MIT (*Cambridge, MA, United States*) Jan. 2025 — Present

- Leveraged CNNs, RL, generative models, and ML force fields for defect identification and materials design
- Co-first-authored three papers in ML-based defect engineering and valence-informed generative material discovery
- Collaborated with the *Quantum Measurement Group* under Prof. Mingda Li

Large Language Model Data Researcher @ ByteDance (*Beijing, China*) Aug. 2025 — Sep. 2025

- Led the design of a benchmark suite of 500+ college-to-PhD-level physics problems for evaluating LLM reasoning
- Contributed to internal projects improving physical reasoning fidelity in frontier LLMs
- Co-developed a test-time scaling (TTS) pipeline enabling gold-medal-level performance on IPhO 2025 theoretical problems

AI Workload Deployment Intern @ Intel Corporation (*Shanghai, China*) Jun. 2025 — Jul. 2025

- Authored internal tutorials on GPU memory layouts (CuTe layout, linear layout) for AI workloads
- Produced technical notes adopted by the CuTeDSL framework team as performance reference materials

Awards & Honors

International Physics Olympiad (IPhO) Jul. 2023
Gold medal, ranked **1st place** (Absolute Winner) in the world

Selected Publications

[†] → Equal contribution; * → Corresponding author

1. Cheng^{†,*}, M., Fu[†], C.-L., **Yu[†], Bowen**, Li*, M. & *et al.* A foundation model for non-destructive defect identification from vibrational spectra. *Matter* **9** (2026).
2. Cheng^{†,*}, M., **Yu[†], Bowen**, Fu, C.-L., Andrejevic, N., Li*, M. & *et al.* Probing Non-Equilibrium Grain Boundary Dynamics with XPCS and Domain-Adaptive Machine Learning. *arXiv preprint arXiv:2605.12194* (2026).
3. Cheng^{†,*}, M., Wan[†], Q., **Yu[†], Bowen**, Li*, M. & *et al.* Reinforcement learning-guided optimization of critical current in high-temperature superconductors. *arXiv preprint arXiv:2510.22424* (2025).

Projects

PaperPlay (HackMIT 2025) Sep. 2025

- In a team of four, built a web platform that converts hand-drawn Mario-style levels into immersive playable games in under 5 minutes, powered by OpenCV and integrated with Modal backend.
- Won **2nd place at HackMIT 2025** in the Modal sponsor track.
- Demo: <https://demo-description.vercel.app/>; Code: <https://github.com/HACKMIT-2025>

Fireroad.ai (MIT CSAIL Agentic AI Hackathon 2026) Apr. 2026

- In a team of four, built an AI-powered Courseroad/Fireroad agent that helps MIT students generate personalized semester plans from academic history, preferences, constraints, and course requirements
- Implemented a React + Node/Express prototype with OpenRouter tool-calling, streaming chat responses, current catalog search, requirement checking, workload summaries, and validated planner UI actions
- Won **MIT CSAIL Agentic AI Hackathon 2026** in Agents for MIT Track; code available at: <https://github.com/bowenyu066/Fireroad.ai>

Skills

- Python, C/C++, PyTorch, NumPy, Triton, CUTLASS, MACE, MLFFs; English (*fluent*), Chinese (*native*)